

AXIOMOS

The runtime that lets robots change safely while they run.

Admission, rollback, and audit for robot fleets.

Updating a robot fleet is a fire drill.

Every behavior change restarts the whole validation loop.

THE LOOP - EVERY BEHAVIOR CHANGE



18

fleet operators interviewed in customer discovery.

12 / 18

call update + revalidation a recurring engineering bottleneck.

3d-3wk

reported validation window after a meaningful behavior change.

Robots receive software [releases](#). They do not safely acquire [capabilities](#).

Metrics from ongoing customer discovery.

Robots are becoming continuously programmable. Their safety model isn't.

01 - SOFTWARE VELOCITY

Behavior is shifting from firmware releases to continuously changing software and learned policies.

RT-2 → π0.5 → GR00T → Helix → generated behaviors

02 - FLEET SCALE

The installed base is already large — and every unit is a place a bad update can land.

4.66M

industrial robots operating globally · +9% YoY (IFR)

03 - ACCOUNTABILITY

Updates increasingly need provenance, rollback, authorization and evidence.

Safety and cybersecurity standards are increasing the burden of proving how robot software is updated, authorized, and recovered.

The faster robot behavior changes, the less viable “test it, trust it, ship it” becomes.

The pain exists before the platform does.

Discovery before deployment: the workflow is already broken.

DISCOVERY FUNNEL

18

fleet operators / robotics teams interviewed

12

report update or revalidation pain

5

active design-partner conversations

2

pilot-scoping conversations

250+ robots represented across interviewed fleets.

IN THEIR WORDS

"A behavior change can trigger days of regression testing because we cannot prove what changed downstream."

"Rollback exists at the software level. Proving the robot is safe after rollback is still manual."

"We can deploy remotely. What we cannot do is establish what the new behavior is allowed to do."

Next milestone: 3-5 paid design partners.

Ship robot updates without revalidating the world.

One operator workflow, from new behavior to safe execution.

INPUT A new behavior or software update arrives for the fleet.

01 VERIFY

Signature

Provenance & resource bounds.

02 ADMIT

Policy

Policy + authority check.

03 RUN

Limits

Execute within explicit bounds.

04 OBSERVE

Audit

Audit + runtime monitoring.

05 ROLLBACK

Known-safe

Restore a known-safe state.

TODAY

Safe deployment + rollback + audit.

TOMORROW

Runtime governance for model-generated behavior.

The missing layer sits between intent and actuation.



Models **propose**. AxiomOS decides what is **admissible**.

- Designed to be vendor-neutral across brains and bodies
- Hardware-backed – an independent controller in the loop
- Enforced in the actuation path, not advisory

The safe path is the only path.

Two chips split capability from enforcement.

CAPABILITY

Raspberry Pi 5 · Linux

Verification · admission · policy ·
runtime.

SAFETY
BOUNDARY

ENFORCEMENT

RP2040-class MCU

Actuation authority · hard limits ·
failsafe · e-stop.

→ Linux is not the safety boundary we want to rely on alone. The capability layer can fail without owning final actuation authority – that lives on an independent deterministic controller.

Running end-to-end today on Pi 5 + RP2040

Provisional patent filed

MEASURED INTERRUPT → PROGRAM PATH

211_{ns}

<0.1% of a 1 kHz control-loop budget. Enforcement fits inside the loop with room to spare.

This isn't a simulation of the product. The runtime already runs.

END-TO-END

Admission → constrained execution
→ rejection → rollback.

HARDWARE

Pi 5 + RP2040

Real two-chip rig.

LATENCY

211ns

Measured path.

PRIOR PRODUCTION SYSTEMS

Deterministic firmware shipped at
Cypherock.

10k+ users

WHAT THIS ROUND PROVES

Production-class robot integration



3-5 paid design partners



Reference deployment



Certification groundwork

Start with fleets already paying the revalidation tax.

IDEAL CUSTOMER

ROS 2 / Linux fleets

Field + inspection robotics

Warehouse / AMR

~10-500 robots

Ship software frequently

THE WEDGE

"Ship updates to your fleet without a revalidation fire drill — with rollback and an audit trail."

MOTION

1 Founder-led discovery

3 One production robot

5 Standardize runtime APIs

2 Paid design partner

4 Expand across fleet

6 Enterprise + certification

EXPANSION 1 robot → 1 fleet → every behavior change

A narrow wedge into a much larger robotics software layer.

Honest numbers, on purpose. We monetize the runtime, not the robot.

BEACHHEAD

ROS 2 / Linux commercial fleets

Teams making frequent software changes and feeling the revalidation tax today.

EXPANSION

Runtime + deployment + safety governance

The layer robots need as behavior velocity and accountability both rise.

SOM · 3 YRS

\$0.5–3M

ARR – 10–50 fleet customers, expanding from fleet-tooling contracts into per-robot runtime licensing.

EXPANSION LOGIC

Tooling contract



Fleet deployment



Per-robot runtime licensing



Certification-backed safety runtime

Everyone adjacent can move toward us. None starts from the same position.

CATEGORY	EXAMPLES	WHAT THEY OWN
Status quo	Internal OTA · regression testing · manual rollback	Deployment plumbing – no runtime admission or hardware-backed enforcement
Brains	NVIDIA · Physical Intelligence · Skild	Intent
Robot OEMs	Figure · 1X · ANYbotics	Body
Observability	Foxglove · Formant · InOrbit	Visibility
Middleware	Apex.AI · RTI · eProxima	Transport / certified build
AxiomOS	–	Runtime admission + actuation governance

NVIDIA BUNDLES IT

Vendor-neutral fleets still need an independent policy boundary.

OEMS BUILD IT

Cross-vendor fleets still need shared governance.

APEX EXTENDS

Certified build-time software and runtime admission solve different parts of the lifecycle.

MOAT TODAY

Founder / systems capability · running architecture · hardware-backed enforcement · provisional IP.

MOAT THAT COMPOUNDS

Certification mappings · deployment evidence · policy integrations · fleet-level runtime APIs.

Land as fleet tooling. Earn the safety premium.

Pricing hypotheses until validated by customers.

NOW

\$10–50K

annual pilot / fleet contract

Design-partner / fleet-tooling pricing for safe OTA, rollback and audit – or per-robot tooling where it fits.

LATER

\$1–3K

/ robot / year

Certification-backed safety runtime – the premium follows certification, which this raise funds.

EVENTUALLY

Capability ecosystem

Certified third-party capabilities and APIs once we own the admission point. Optionality, not the current case.

Built by someone who has already shipped where software meets hardware.

Cypherock

PRODUCTION FIRMWARE

Deterministic, security-critical systems.

10k+ users

AxiomOS

KERNEL / RUNTIME

Pi 5 + RP2040 end-to-end system.

211ns

ResQTerra

AUTONOMY UNDER FAILURE

GPS-denied drone · ROS / MAVLink / Jetson.

Patent filed

Polymath

SOFTWARE ENGINEER · YC W26

Simulation & verification environments for long-horizon agents.

Sim / verify

NOW

Runtime working end-to-end.

0-6 MONTHS

Production robot + first paid design partners.

6-12 MONTHS

3-5 paid design partners + runtime APIs.

12-18 MONTHS

Reference deployments + certification mapping + early ARR.

Make changing a robot safer than reflashing one.

\$1.5M pre-seed

18-24 months of runway.

2 senior engineering hires

Production robot integration

Certification groundwork

Design-partner deployments

MILESTONE → SEED

3-5 paid design partners + reference deployments.

The brain is getting cheap. The trust layer is getting essential.